



2.5

## Exponential Function Context and Data Modeling

### Core-to-Challenge Worksheet and AP Practice

AP Precalculus - Unit 2 | ECHS Mathematics | Teacher: Mohammad Abu-Ghuwaleh

#### Student Information

Name: \_\_\_\_\_ Class: \_\_\_\_\_ Date: \_\_\_\_\_

#### Directions

Show mathematical evidence. Retain domain restrictions, label graphs, include units, and write complete interpretations. Calculator use is permitted only when the item explicitly requires approximation, regression, or numerical solving.

#### Readiness

1. Write one precise sentence explaining this principle: Initial value is the output at the chosen time origin.  
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2. Write one precise sentence explaining this principle: A rate of  $r$  per period gives factor  $1 + r$  or  $1 - r$ .  
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3. Write one precise sentence explaining this principle: Two-point data determine the factor through a root when inputs differ by more than 1.  
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4. Write one precise sentence explaining this principle: Predictions require appropriate units and a stated domain.  
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**Core**

5. A culture starts at 500 cells and grows 18% per hour. Write a model.
6. A value falls from 80 to 50 in 6 years. Find the annual factor.
7. Interpret  $A(t) = 240e^{-0.07t}$ .
8. State what the formula  $A(t) = A_0(1 + r)^t$  tells you and name any required restriction.
9. State what the formula  $A(t) = A_0e^{kt}$  tells you and name any required restriction.
10. State what the formula  $k = \frac{\ln(A(t)/A_0)}{t}$  tells you and name any required restriction.

**Medium**

11. Write an exponential model: initial 900, growth 6% yearly.
12. Write an exponential model: initial 240, decay 18% monthly.
13. Write an exponential model: initial 70, doubles every 5 units.
14. Write an exponential model: initial 1200, half-life 12 years.

15. Write an exponential model: initial 15, continuous rate 0.08.

**Hard**

16. A population is 5000 at  $t=2$  and 7200 at  $t=6$ . Construct a model centered at  $t=2$ .

17. For  $A = 300(0.91)^t$ , find the first integer  $t$  with  $A < 100$ .

18. Interpret every parameter in  $T(t) = 18 + 72e^{-0.25t}$ .

19. Convert 2.4% monthly growth to an equivalent annual factor.

**Difficult**

20. A quantity grows from 40 to 100 in 7 years. Find doubling time.

21. Find continuous rate  $k$  equivalent to 9% annual growth.

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22. A model  $A = ab^t$  satisfies  $A(3)=24$  and  $A(8)=6$ . Find  $a$  and  $b$ .

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### Challenge / HOT

23. Explain why a negative exponential prediction is impossible for  $ab^x$  when  $a>0, b>0$  and what a negative observation implies.

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24. Construct two equivalent models using different time units for a 30% increase every 5 years.

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### AP-Style Multiple Choice

25. 18% decay factor is

- (A) 0.18  
(B) 0.82  
(C) 1.18

(D) -0.18

26. Doubling time  $d$  gives model factor

(A)  $2^t$

(B)  $2^{t/d}$

(C)  $d^{2t}$

(D)  $(1 + d)^t$

27. Continuous model uses

(A)  $e^{kt}$

(B)  $kt$

(C)  $\ln t$

(D)  $t^k$

28. If  $b < 1$ , model shows

(A) growth

(B) decay

(C) linear change

(D) undefined behavior

29. Equivalent annual factor from monthly  $b$  is

(A)  $12b$

(B)  $b/12$

(C)  $b^{12}$

(D)  $b + 12$

30. Solving for time usually requires

(A) factoring only

(B) a logarithm

(C) a derivative

(D) a matrix

### AP-Style Free Response

31. A city population is 85,000 in 2020 and 103,000 in 2026. Assume exponential growth. (a) Construct model with  $t$  years after 2020. (b) Predict 2035. (c) Find when population reaches 150,000. (d) State limitation.

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32. A medication level follows  $M(t) = D + Ae^{-kt}$ . Measurements are  $M(0)=90$ , long-run level 10, and  $M(4)=30$ . Find  $A, k$  and predict  $M(8)$ .

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**Reflection**

Identify one item where a domain restriction, unit, assumption, or representation changed your reasoning. Explain in 2-3 sentences.

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Original ECHS questions. Selected concepts are informed by Pearson and Holt Precalculus; no textbook exercises are reproduced.